

Package #160

Install low■drag helical rib turbulators (smooth■walled, optimized pitch and height) inside the tube, combined with a passive pulsating flow pattern (alternating flow direction every few minutes) to increase shear without adding significant pressure loss.

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented problem 'fouling' concerning passive countercurrent heat exchanger. User-submitted engineering problem (session input): "Independent audit case HX-2026-0910-B: Investigate whether a passive countercurrent heat exchanger can recover at least 60 percent of sensible heat from 40 C industrial wastewater at 2 L/min into 15 C clean inlet water while maintaining less than 10 kPa pressure drop, with fouling-prone suspended so". ROUTING: doe_osti: routing state SUSPENDED_RELEVANCE — measured relevance instability (R375: same fusion papers relevant 3 -> 0; pooled relevant-rate 0.175 pre-fix; DETERMINISM_GAP flagged in the R394 production audit) — records poison synthesis-feeding retrieval more than they feed it until relevance is semantically adjudicated; reinstatement: the Phase P1 semantic reranker lands (semantic relevance adjudication between query and record, replacing lexical overlap); flip to ACTIVE + a measured relevance re-run + worklog entry. [fra_rail_accidents] retrieval status GRAMMAR_MISMATCH (query does not match source grammar: YYYY-MM-DD date floor (\$where date >= X) — engine routing error, not a provider failure (not absence, Art. XXV)) — provider outcome recorded, not treated as absence (Art. XXI.3). [doe_osti] retrieval status NOT_QUERIED_SUSPENDED_RELEVANCE (engine routing decision (R399 W3): SUSPENDED_RELEVANCE — not a provider failure and not absence) — provider outcome recorded, not treated as absence (Art. XXI.3). Problem-existence basis: NO SOURCE RETURNED RECORDS — problem existence UNVERIFIED by live evidence (Art. XX); treat as hypothesis.

Mechanism: Vortex■induced shear from helical turbulator ribs creates secondary flows that continuously dislodge particles, reducing fouling while enhancing convective heat transfer, allowing higher heat recovery at low pressure drop.

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Status: ENGINEERING_DEFINITION. Not physically validated.